

# LUC Blassel

Researcher - Machine Learning for Biology and Evolution

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📍 Paris, France

Passionate about machine learning applied to biological problems, I did a PhD at the Institut Pasteur in Paris. I am currently a post-doctoral researcher using deep attention networks to study evolution.

## Experience

**Post-Doctoral Researcher**, [Sorbonne Université](#), Paris, France

2023–

During this postdoc I will work on the [PhyloFormer](#) project with [Laurent Jacob](#) and members of the LBBE in Lyon. I am working on improving performance of the model as well as its generalisation. Extensions to the model are planned, such as taking unaligned sequences as input and including a differentiable tree-building step in an end-to-end pipeline.

**Doctoral candidate**, [Institut Pasteur](#), Paris, France

2020–2023

PhD in computational biology, focused on improving and learning from sequence alignments. With the ever growing quantity of high quality sequencing data, we are now able to develop methods to take advantage of this data. My PhD project followed 2 different axes: (1) Using machine learning to discover Drug resistance mutations in HIV (Continuation of the M.Sc project) (2) Search of a function space for sequence transformation functions that improve sequence mapping and alignment.

This PhD was supervised by [Rayan Chikhi](#) and funded by the [PRAIRIE](#) institute. The manuscript can be found [here](#)

**Research Engineer**, [Institut Pasteur](#), Paris, France

2018–2020

Using machine learning and large datasets to explore the drug resistance landscape in HIV. Supervised by [Olivier Gascuel](#). A large dataset of UK HIV sequences associated with patient RT-treatment status was used to train several machine learning algorithms. Examining the important learned features gave us insight in mutations associated to ART failure. This project resulted in a [publication](#). Project page [here](#), and [M.Sc thesis](#) (in french).

**Data Scientist Intern**, [CapitalData](#), Paris, France

Mar-Sept 2017

As a data scientist intern at Capital data I was tasked with developing a software pipeline to validate data prior to database integration, and detect anomalies using time series. This was implemented in R and Python. I also, developed reporting-oriented data visualizations of KPIs using BIME and Tableau. Finally statistical AB-testing was also done using python.

**Teaching and tutoring**, self-employed, Paris, France

2015–2022

Since the beginning of my higher education, I have been tutoring middle school, high school and university students in scientific subjects, mainly mathematics. The aim is to turn the students into independent people, more confident in their abilities and more knowledgeable in order to excel in their respective studies. Since September 2021, I have been doing oral examination in English for PCSI preparatory class students at the [ENCPB](#) in Paris, France.

## Education

**Doctorate (PhD)**, [Sorbonne Université](#), Paris, France

2020–2022

Worked at the Institut Pasteur on machine learning techniques applied to biological sequence alignments.

**Masters of Science (MSc)**, [Dauphine université](#), Paris, France

2017–2018

during the [IASD](#) program, I followed courses in Deep learning, reinforcement learning, statistical modelling, software engineering and data science. This was a double curriculum with the AgroParisTech IODAA program.

**Diplôme d'Ingénieur**, [AgroParisTech](#), Paris, France

2014–2018

The first two years of this program are focused on biology with courses in molecular biology, ecology, health, epidemiology and statistics.

During the final year, I followed the [IODAA](#) program, which was done as a double curriculum with Dauphine université, studying machine learning, data science and statistical modelling with a focus on biological problems.

BCPST Preparatory class, École Saint-Hilaire, Paris, France

2012–2014

2 year French preparatory class to prepare a nationally ranked exam for entry in the "Grande Écoles"  
ranked 201/1967 (Veto), ranked 481/2991 (Agro)

## Languages

Mother tongue

Other languages<sup>1</sup>

German<sup>2</sup>

Mandarin Chinese<sup>3</sup>

### French & English

| Understanding  |                | Speaking       |                | Writing        |
|----------------|----------------|----------------|----------------|----------------|
| Listening      | Reading        | Interaction    | Production     |                |
| B1 Independent | B1 Independent | B1 Independent | B1 Independent | B1 Independent |
| A2 Basic       | A2 Basic       | A2 Basic       | A2 Basic       | A2 Basic       |

<sup>1</sup> Common European Framework of Reference for Languages (CEFR)

<sup>2</sup> 1 month program at **Goethe Institute** in Schwäbisch Hall, Germany

<sup>3</sup> 5 month program at **SISU** Shanghai, Chian & 1 month program at UIR, Beijing, CN

## Publications (Google scholar, ORCID)

<sup>f</sup> co-first authors

### Journal articles

Phyloformer: Fast, accurate and versatile phylogenetic reconstruction with deep neural networks

Luca Nesterenko <sup>f</sup>, **Luc Blassel** <sup>f</sup>, Philippe Veber, Bastien Boussau, Laurent Jacob

*Molecular Biology and Evolution*, msaf051, **2025**, [10.1093/molbev/msaf051](https://doi.org/10.1093/molbev/msaf051)

Mapping-friendly sequence reductions: Going beyond homopolymer compression

**Luc Blassel**, Paul Medvedev, Rayan Chikhi

*iScience* 25.11, p. 105305, **2022**, [10.1016/j.isci.2022.105305](https://doi.org/10.1016/j.isci.2022.105305)

Using Machine Learning and Big Data to Explore the Drug Resistance Landscape in HIV

**Luc Blassel**, Anna Tostevin, Christian Julian Villabona-Arenas, Martine Peeters, Stéphane Hué, Olivier Gascuel

*PLOS Computational Biology* 17.8, e1008873, **2021**, [10.1371/journal.pcbi.1008873](https://doi.org/10.1371/journal.pcbi.1008873)

Drug Resistance Mutations in HIV: New Bioinformatics Approaches and Challenges

**Luc Blassel** <sup>f</sup>, Anna Zhukova <sup>f</sup>, Christian J Villabona-Arenas, Katherine E Atkins, Stéphane Hué, Olivier Gascuel

*Current Opinion in Virology* 51, pp. 56–64, **2021**, [10.1016/j.coviro.2021.09.009](https://doi.org/10.1016/j.coviro.2021.09.009)

Origin, Evolution and Global Spread of SARS-CoV-2

Anna Zhukova, **Luc Blassel**, Frédéric Lemoine, Marie Morel, Jakub Voznica, Olivier Gascuel

*Comptes Rendus. Biologies* 344.1, pp. 57–75, **2021**, [10.5802/crbior.29](https://doi.org/10.5802/crbior.29)

COVID-Align: accurate online alignment of hCoV-19 genomes using a profile HMM

Frédéric Lemoine, **Luc Blassel**, Jakub Voznica, Olivier Gascuel

*Bioinformatics* (btaa871), **2020**, [10.1093/bioinformatics/btaa871](https://doi.org/10.1093/bioinformatics/btaa871)

### Preprints & Submitted articles

Teddy: Neural Inference of Epidemiological Parameters from Viral Sequences

Vincent Garot, **Luc Blassel**, Luca Nesterenko, Anna Zhukova, Samuel Alizon, Laurent Jacob

*bioRxiv*, p. 2026.01.05.697728, **2026**, [10.64898/2026.01.05.697728](https://doi.org/10.64898/2026.01.05.697728)

Likelihood-free inference of phylogenetic tree posterior distributions

**Luc Blassel**, Bastien Boussau, Nicolas Lartillot, Laurent Jacob

*arXiv*, **2025**, [10.48550/arXiv.2510.12976](https://arxiv.org/abs/2510.12976)

### Conference talks & Posters

Deep end-to-end likelihood-free inference of phylogenetic trees

**Luc Blassel**, Bastien Boussau, Nicolas Lartillot, Laurent Jacob

*MLCB*, 2025

Deep likelihood-free inference of phylogenetic trees

**Luc Blassel**, Bastien Boussau, Nicolas Lartillot, Laurent Jacob

*MASAMB*, 2025

Likelihood-free inference of phylogenetic tree posterior distributions

**Luc Blassel**, Bastien Boussau, Nicolas Lartillot, Laurent Jacob

*LEGO*, 2025

Likelihood-free inference of phylogenetic tree posterior distributions

**Luc Blassel**, Bastien Boussau, Nicolas Lartillot, Laurent Jacob

*LEGEND*, 2025

Phyloformer: Fast, accurate and versatile phylogenetic reconstruction with deep neural networks

Luca Nesterenko <sup>f</sup>, **Luc Blassel** <sup>f</sup>, Philippe Veber, Bastien Boussau, Laurent Jacob

*MCEB*, 2024

Mapping-Friendly Sequence Reductions: Going beyond Homopolymer Compression

**Luc Blassel**, Paul Medvedev, Rayan Chikhi

*RECOMB-SEQ*, 2022

Machine learning approaches to reveal resistance mutations in HIV

Luc Blassel, Anna Zhukova, Olivier Gascuel

*MCEB*, 2019, (poster)